

ZARI.ai - Master Project & Technical Report

End-to-End 3-Crop Disease Detection, EDL Uncertainty, RAG Advisory & MLOps System

1. Executive Summary & System Overview

ZARI.ai is an end-to-end plant disease detection and context-aware recommendation system engineered specifically for Tomato, Potato, and Bell Pepper crops in Pakistan.

* Final System Status: PRODUCTION_READY_WITH_LIMITATIONS

* Real CUDA

2. Complete Repository Directory & File Structure

* backend/main.py: FastAPI backend web server for serving real-time inference requests.

* ml_pipel
class_aliases

3. Core Technical Methodologies & Algorithms Explained

1. Evidential Deep Learning (EDL) & Dirichlet Uncertainty:

Standard Softmax over crop images. EDL uses Dirichlet priors: $\sum(e_k + 1)$

2. Selective Classification and Risk Control (SCRC):

Calibrated quality scores. Confidence is adjusted for risk.

3. SAM2 Leaf Mask Segmentation:

Uses central point and leaf contours for precise segmentation.

4. Grad-CAM Heatmap Localization & Visual Severity Proxy:

Grad-CAM activation maps. Localization accuracy: 85-95% (15-35%), and severity proxy.

5. Multilingual RAG Vector Database:

Dense multilingual embeddings (208 chunks a

6. IPM Advisory Rules & Active-Ingredient Chemical Policy:

Enforces strict regulatory compliance. Only (Mancozeb)

7. Pathogen Weather Risk Heuristics:

Injects environmental data into visual disease models.

8. ViT vs CNN Explainability (Swin-Tiny vs EfficientNetV2-B2):

Swin-Tiny ViT vs EfficientNetV2-B2 CNN explainability.

4. Complete System Parameter Breakdown

* Model A (Crop Router - EfficientNetV2-B2): 7,705,221 parameters (7.71 M)

* Model B Torque Sensor (Swin-Tiny ViT): 21,811,264 parameters (21.81 M)

5. Strategic Operational Decision Framework

1. Vision Backbone: Keep EfficientNetV2-B2 locked for v1 (Native Grad-CAM explainability).

2. Advisory Engine: Integrate RAG and heuristics for real-time IPM advice.

6. Per-Crop AUROC & SCRC Calibrated Threshold Metrics

Crop	Test Samples	Accuracy	Macro F1	Test AUROC	SCRC Threshold (u)	Coverage	Selective Risk	SCRC FAR		
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7. Epoch-by-Epoch Training & Validation Error Tables

A. Model A Crop Router (13 Epochs)

Epoch | Stage

B. Model B Tomato Classifier (13 Epochs)

Epoch | Stage

C. Knowledge Distillation Student (5 Epochs)

Epoch | Total

8. Knowledge Distillation & Swin Comparison Results Matrix

Metric	Production (Model B EfficientNet)	Distilled (EfficientNet Student)	Change		
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9. Pashto Multilingual Retrieval Benchmark Results

* Test Query: " " (Tomato Late Blight Treatment in Pashto)

* Before Scor